

# Blood Transfusion in Obstetrics and Gynaecology

## Contents

|                                                                 |          |
|-----------------------------------------------------------------|----------|
| <b>Blood Transfusion in Obstetrics &amp; Gynaecology</b>        | <b>1</b> |
| Outline . . . . .                                               | 1        |
| Introduction . . . . .                                          | 1        |
| Component therapy – the way forward . . . . .                   | 2        |
| Indications for transfusion . . . . .                           | 2        |
| Blood components and their indications . . . . .                | 2        |
| Types of transfusion . . . . .                                  | 2        |
| Exchange blood transfusion (EBT) . . . . .                      | 3        |
| Autologous blood transfusion . . . . .                          | 3        |
| Preoperative autologous donation . . . . .                      | 3        |
| Intraoperative cell salvage (IOCS) . . . . .                    | 3        |
| Acute normovolaemic haemodilution . . . . .                     | 3        |
| General principles . . . . .                                    | 3        |
| Administration . . . . .                                        | 4        |
| Adverse effects . . . . .                                       | 4        |
| Patient Blood Management (PBM) . . . . .                        | 4        |
| Managing Jehovah’s Witnesses (refusal of transfusion) . . . . . | 4        |
| References . . . . .                                            | 5        |

## Blood Transfusion in Obstetrics & Gynaecology

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### Outline

Introduction · Component therapy · Indications · Blood components & their uses · Types of transfusion · Exchange & autologous transfusion · Intraoperative cell salvage · Acute normovolaemic haemodilution · General principles · Administration · Adverse effects · Patient Blood Management · Jehovah’s Witnesses

### Introduction

Blood transfusion remains one of the most important treatment modalities in Obstetrics & Gynaecology, especially in **tropical obstetrics, where haemorrhage is the number one cause of maternal mortality and morbidity**.

With **HIV/AIDS and the difficulty recruiting voluntary donors**, transfusion should be used **only to save the mother’s life**.

## Component therapy – the way forward

Technological advances allow a unit of **whole blood** to be separated into **components** – erythrocytes plus one or two of **platelets, cryoprecipitate, or fresh frozen plasma (FFP)**.

**Component therapy is now the standard** and has made whole blood rarely necessary because it:

- Gives **only the specific component needed** -> better treatment
- **Conserves blood** – one donated unit can treat **several patients**

Transfusion is often criticised in hindsight as “**too little, too late.**” Women at high risk of losing **> 1000 ml** should be **strongly advised to deliver where transfusion and intensive care are available.**

Transfusion is life-saving but not without risk: **transfusion-transmitted infections**, immunological sequelae (e.g. **red-cell alloimmunisation**), and – the **major risk** – receiving an “**incorrect blood component**”. Hence **strict adherence to correct sampling, cross-match and administration** is paramount, even in emergency.

## Indications for transfusion

- **Symptomatic anaemia**, including heart failure
- **Very low Hb** – less than **5.6 g/dl**
- **Anaemia very close to term**
- **Anaemia in labour**
- **Acute haemorrhage / operative blood loss > 30% of blood volume or > 500 ml** where tissue oxygenation is not maintained by plasma expanders

## Blood components and their indications

| Product                      | Content                                                      | Indication                                                                            |
|------------------------------|--------------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>Red blood cells</b>       | Red cells                                                    | Increase oxygen-carrying capacity in anaemia; orthostatic hypotension from blood loss |
| <b>Platelet concentrates</b> | Platelets                                                    | Control/prevent bleeding from deficiencies in platelet number or function             |
| <b>Fresh frozen plasma</b>   | Plasma + clotting factors                                    | Increase clotting factors in demonstrated deficiency                                  |
| <b>Cryoprecipitate</b>       | Factors I, V, VIII, XIII, von Willebrand factor, fibronectin | Deficiency of fibrinogen, factor VIII, factor XIII, fibronectin or vWF                |

## Types of transfusion

- **Whole blood** transfusion
- **Packed cells** (sedimented blood)
- **Blood component therapy**
- **Exchange blood transfusion**
- **Autologous blood transfusion**

## Exchange blood transfusion (EBT)

The **most rapid** method of correcting anaemia, but **most expensive and laborious**. Advantage over packed cells is only **marginal** and greatest in the **first 12 hours**. Now **rarely used**.

The only indication for EBT in anaemia of pregnancy is when **correction must be made very quickly** – e.g. a patient with **severe anaemia in early labour**.

## Autologous blood transfusion

The **collection and reinfusion of the patient's own blood**. Three techniques:

1. **Preoperative autologous blood donation**
2. **Intraoperative blood salvage**
3. **Acute normovolaemic haemodilution**

### Preoperative autologous donation

- **Donate at least 2 weeks** before surgery/delivery; the patient must have **Hb  $\geq 11$  g/dl**
- **Pre-delivery autologous deposit is NOT recommended** – few parturients need transfusion, so routine donation isn't justified
- **Placenta praevia** is one obstetric condition for which it may be appropriate
- Risks are minor – **vasovagal reactions in 2-5%**
- Indications are the **same as for RBC transfusion**

### Intraoperative cell salvage (IOCS)

The **sterile collection and reinfusion of shed blood**.

- **Contraindications:** infection, contamination with **malignant cells**, penetrating bowel wounds
- Uses a **continuous-flow centrifugation device** (e.g. **Cell Saver**) that **washes** cells as collected – the old “scoop and sieve through gauze” method is **no longer acceptable**
- Salvaged blood must **not** be transfused to another patient, must be properly identified, and if stored, kept  **$\leq 24$  h** (room temperature  **$\leq 4$  h**; best not to leave theatre/recovery)
- Recommended where anticipated blood loss is  **$> 20\%$  of estimated blood volume**; current evidence supports its use in obstetrics, though large RCTs are still needed

### Acute normovolaemic haemodilution

- **Removal of blood immediately before/after induction of anaesthesia**, with **crystalloid given simultaneously** to keep normovolaemia; the patient's RBCs are **reinfused at the end**
- Usually performed by the **anaesthetist**
- **Fewer RBCs are lost** during surgery (lower Hb concentration), plus fresh whole blood is available and **lower viscosity improves tissue perfusion**

## General principles

- Obtain **valid consent** where possible; if not feasible in an emergency, provide information **retrospectively** and **document** the reason and consent discussion
- All women should have **blood group and antibody status checked at booking and at 28 weeks**
- Transfuse **ABO, RhD and K (Kell) compatible** red cells
- Provide **CMV-seronegative** red cell/platelet components for elective transfusion in pregnancy

- Major obstetric haemorrhage protocols must provide **emergency O RhD-negative, K-negative** units immediately, switching to **group-specific** blood as soon as feasible

## Administration

- **Correct identification of each unit is critical** – often a **second person cross-checks** before transfusion
- **Warming** is rarely needed except when rate  $> 50 \text{ ml/kg/hr}$  (risk of transfusion-induced cardiac hypothermia) or in **cold agglutinin disease**
- Administer through a **filter** to remove debris; **leucocyte filters** prevent **febrile non-haemolytic reactions** and lower CMV risk (*but an RBC leucocyte-depleting filter used for platelets removes most platelets*)
- **Only normal saline** may share the line – other crystalloids cause **agglutination/haemolysis**; **calcium-containing solutions cause clotting** in the tubing
- **No medications** should be added to blood – interactions are unpredictable, reactions become hard to attribute, and stopping the transfusion means an incomplete drug dose

## Adverse effects

Major complications: **infections, alloimmunisation and transfusion reactions.**

- **Infections** – HIV/AIDS, hepatitis, and **CMV** (a threat to the immunocompromised); **CMV-negative blood for all intrauterine transfusions and undelivered parturients**
- **Acute haemolytic reactions** – mostly from **ABO incompatibility** -> **intravascular haemolysis**. Signs: **chills, fever, chest and flank pain, nausea, cardiovascular collapse, and DIC**
- **Febrile non-haemolytic reactions** – recipient antibodies against leucocyte/platelet antigens; respond to **antipyretics**; use **leucocyte-reduced** components if recurrent
- **Delayed haemolytic reactions** – from **Kidd (Jk), Duffy (Fy), Kell** groups; usually **extravascular**, with fever, anaemia, hyperbilirubinaemia and a **positive direct Coombs test**, occurring **~7-10 days** after transfusion
- Less serious: allergic reactions, febrile reactions, **circulatory overload**

## Patient Blood Management (PBM)

An **all-encompassing approach to avoid unnecessary transfusion**, recommended by the AABB and others:

- **Optimise pre-operative/pre-delivery Hb**
- **Avoid over-transfusion**
- **Cell salvage** where appropriate
- Accept **evidence-based lower transfusion triggers**
- **Iron supplements** (IV or oral) for women not actively bleeding and cardiovascularly stable

## Managing Jehovah's Witnesses (refusal of transfusion)

- **Careful multidisciplinary planning** with senior input to minimise anaemia and manage blood loss
- **Early iron replacement**, including IV iron if needed
- **Discuss and document consent/refusal** and transfusion-sparing techniques **antenatally**
- Consider **pharmacological, mechanical and surgical** measures early to avoid banked blood

- Ensure the woman is **fully informed of the risks of refusal** and maternal mortality data – handled **sensitively, without coercion**

## References

Standard O&G texts; RCOG guidance on blood transfusion in obstetrics; lecture material of Prof E O Ezirim.